BI project - Maven Market

In this fictional BI project, I step into the role of a consultant, solving a business question for a grocery retail company by developing an interactive dashboard

Tools used: MySQL, Power BI (Power Query, DAX)

The project was developed in close collaboration with a simulated client, ensuring that all steps align with business requirements and stakeholder needs

Data Extraction

Data was extracted from a MySQL database to simulate a realistic business environment and was structured into a practical star schema for analysis

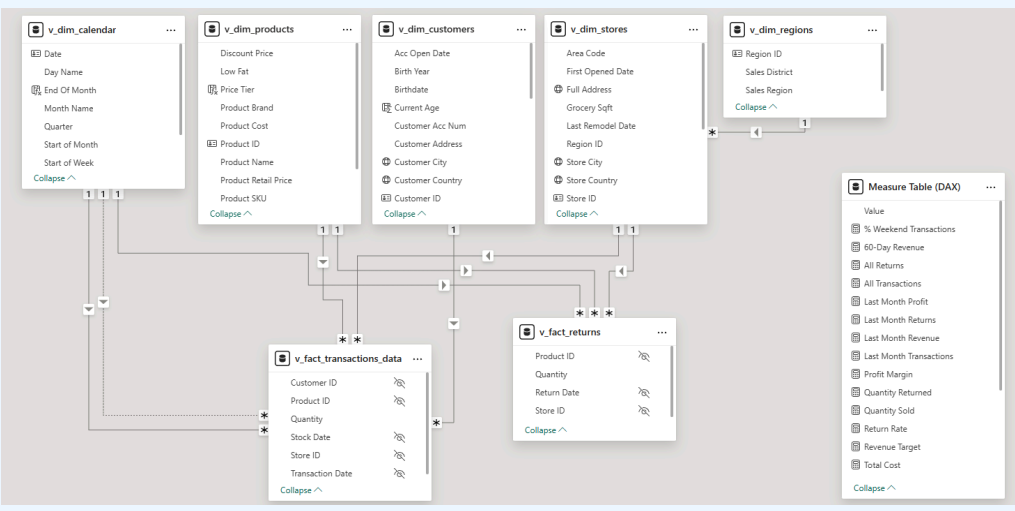
```sql
USE p2_mavenmarket;

CREATE VIEW v_fact_transactions_data AS
SELECT
t97.customer_id,
t97.product_id,
t97.quantity,
t97.stock_date,
t97.store_id,
t97.transaction_date
FROM MavenMarket_Transactions_1997 t97
UNION ALL
SELECT
t98.customer_id,
t98.product_id,
t98.quantity,
t98.stock_date,
t98.store_id,
t98.transaction_date
FROM MavenMarket_Transactions_1998 t98;

CREATE VIEW v_fact_returns AS
SELECT
product_id,
```

sql/SQLViews_Query *Data Source*

Shaping Data & Creating Data Model



Data was transformed in Power Query, where additional columns were created to support analytical measures. Relationships between tables were defined in the model view to ensure a well-structured and reliable data model

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DAX Measures Development

```
Weekend Transactions =
CALCULATE(
    COUNTROWS('v_fact_transactions_data'),
    'v_dim_calendar'[Weekend] = "Y"
)
```

```
Profit Margin =
[Total Profit] / [Total Revenue]
```

```
Last Month Transactions =
CALCULATE(
    [Total Transactions],
    DATEADD(
        'v_dim_calendar'[Date],
        -1,
        MONTH
    )
)
```

```
60-Day Revenue =
CALCULATE(
    [Total Revenue],
    DATESINPERIOD(
        'v_dim_calendar'[Date],
        MAX(
            'v_dim_calendar'[Date]
            ),
            -60,
            DAY
    )
)
```

```
Revenue Target =
[Last Month Revenue] * 1.05
```

DAX was used to create key measures such as revenue, transactions, and time-based calculations, enabling dynamic and interactive dashboard reporting

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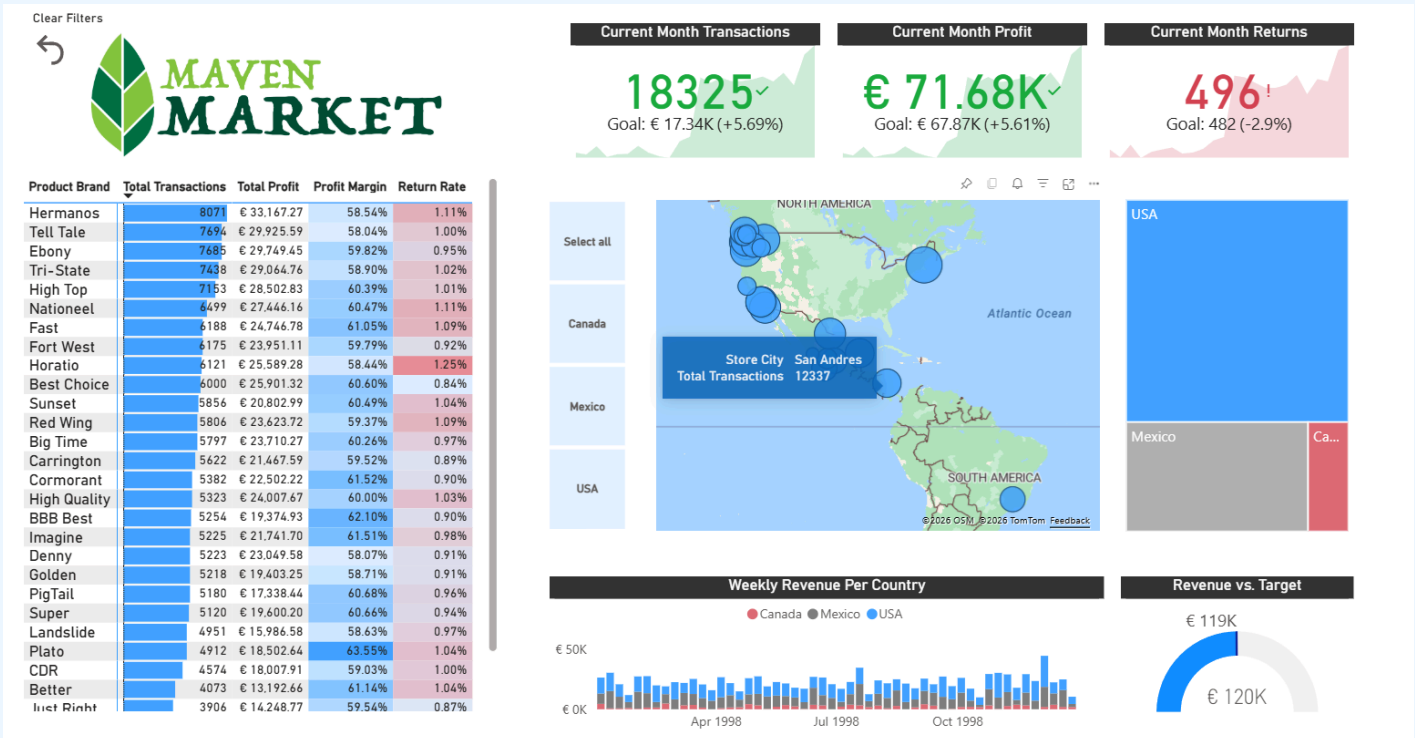
Building The Report

The final dashboard aligns with the client's requirements and is designed to effectively communicate key insights



Product Brand	Total Transactions	Total Profit	Profit Margin	Return Rate
Hermanos	8071	€ 33,167.27	58.54%	1.11%
Tell Tale	7694	€ 29,925.59	58.04%	1.00%
Ebony	7685	€ 29,749.45	59.82%	0.95%
Tri-State	7438	€ 29,064.76	58.90%	1.02%
High Top	7153	€ 28,502.83	60.39%	1.01%
Nationeel	6499	€ 27,446.16	60.47%	1.11%
Fast	6188	€ 24,746.78	61.05%	1.09%
Fort West	6175	€ 23,951.11	59.79%	0.92%
Horatio	6121	€ 25,589.28	58.44%	1.25%
Best Choice	6000	€ 25,901.32	60.60%	0.84%
Sunset	5856	€ 20,802.99	60.49%	1.04%
Red Wing	5806	€ 23,623.72	59.37%	1.09%
Big Time	5797	€ 23,710.27	60.26%	0.97%
Carrington	5622	€ 21,467.59	59.52%	0.89%
Cormorant	5382	€ 22,502.22	61.52%	0.90%
High Quality	5323	€ 24,007.67	60.00%	1.03%
BBB Best	5254	€ 19,374.93	62.10%	0.90%
Imagine	5225	€ 21,741.70	61.51%	0.98%
Denny	5223	€ 23,049.58	58.07%	0.91%
Golden	5218	€ 19,403.25	58.71%	0.91%
PigTail	5180	€ 17,338.44	60.68%	0.96%
Super	5120	€ 19,600.20	60.66%	0.94%
Landslide	4951	€ 15,986.58	58.63%	0.97%
Plato	4912	€ 18,502.64	63.55%	1.04%
CDR	4574	€ 18,007.91	59.03%	1.00%
Better	4073	€ 13,192.66	61.14%	1.04%
Just Right	3906	€ 14,248.77	59.54%	0.87%

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images/dashboard.png

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April 2026